

Cavity-induced persistent current in a π loop

Model. We consider a superconducting (SC) loop of L sites closed by an interacting quantum dot (site 0) and threaded by the magnetic flux of a single lumped LC mode [Fig. 1(a)]. The full microscopic Hamiltonian is

$$H = \hbar\omega a^\dagger a - t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c_{r+1\sigma}^\dagger c_{r\sigma} + \text{h.c.} \right) + \Delta \sum_{r=1}^{L-1} \left(c_{r\uparrow}^\dagger c_{r\downarrow}^\dagger + \text{h.c.} \right) + \epsilon_d n_0 + U n_{0\uparrow} n_{0\downarrow} - t_J \sum_{\sigma} \left(c_{0\sigma}^\dagger c_{L-1\sigma} + \text{h.c.} \right) - t_J \sum_{\sigma} \left(e^{i\varphi_{\text{cl}}} e^{ig(a+a^\dagger)} c_{1\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right). \quad (1)$$

The first line is the resonator and an s -wave BCS chain (hopping t , pairing Δ); the second line is the Anderson impurity (level $\epsilon_d = -U/2$, on-site repulsion U) tunnel-coupled (t_J) to the two ends of the chain; the third line carries the total Peierls phase of the loop, which by a gauge choice is placed on a single bond. The vector potential of the mode enters through the gauge-invariant displacement operator $D = e^{ig(a+a^\dagger)}$, so diamagnetic (A^2) terms are included to all orders; $g = 2\pi\Phi_{\text{zpf}}/(h/e)$ is the electron phase generated by the zero-point flux of the mode, and φ_{cl} is a static external flux. An electron phase $\varphi = \pi$ corresponds to one superconducting flux quantum $\Phi_0 = h/2e$. For a lumped resonator of impedance Z , galvanically sharing its inductance with the loop, $(2g)^2 = \pi Z/R_Q$ with $R_Q = h/4e^2$. Unless stated otherwise we take $t = 1$, $\Delta = t$, $t_J = 0.5t$, $L = 14$ ($L \gg \xi$), $\hbar\omega = 7.6 \times 10^{-4}t$ and *no* external flux, $\varphi_{\text{cl}} \rightarrow 0^+$ (a seed $\varphi_{\text{cl}} = 10^{-3}$ is used to select one of the degenerate states). Two auxiliary versions are also used: a weak link ($U = 0$, dot replaced by a bond t_J) biased by $\Phi_0/2$, and the short ring with a Zeeman-polarized dot of Ref. [1]. We write Eq. (1) compactly as $H = \hbar\omega a^\dagger a + H_{\text{ring}}[\varphi_{\text{cl}} + g(a + a^\dagger)]$.

The only property of H_{ring} that enters below is its ground-state energy at fixed phase,

$$E(\varphi) = \sum_{m \geq 0} E_m \cos m\varphi. \quad (2)$$

Even harmonics ($m = 2, 4, \dots$) describe Cooper-pair tunnelling ($h/2e$ periodicity). The $m = 1$ term is the single-electron (h/e) persistent current of a mesoscopic ring [2] and is suppressed as $e^{-L/\xi}$. A 0-junction has $E_2 < 0$ and a π -junction has $E_2 > 0$. When a 0-junction is biased by $\Phi_0/2$, $\varphi_{\text{cl}} = \pi/2$ maps $E_2 \cos 2\varphi \rightarrow -E_2 \cos 2\varphi$. The two routes to a π loop are therefore equivalent up to the odd harmonics, which turn into an explicit symmetry-breaking term $-E_1 \sin \varphi$. In contrast, an intrinsic π -junction keeps $E(\varphi) = E(-\varphi)$ exactly, since $H_{\text{ring}}[0]$ is real (time-reversal symmetry).

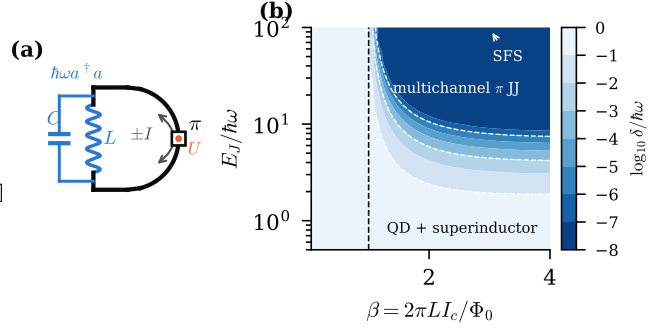


FIG. 1. (a) A superconducting loop closed by a π junction (here an interacting dot with charging energy U) shares its inductance L with an LC mode; the two degenerate states carry opposite persistent currents $\pm I$. (b) Tunnel splitting δ between the two current-carrying states in the plane of the screening parameter β and the quantum parameter $E_J/\hbar\omega$ [Eq. (3) with the DMRG $E(\varphi)$ of the $U = 2t$ junction]. White lines: $\delta/\hbar\omega = 10^{-6}, 10^{-3}, 10^{-1}$; dashed: classical threshold $\beta = 1$. Labels: typical regimes of the platforms discussed in the text.

Adiabatic reduction and Landau theory. For $\hbar\omega$ well below the Andreev gap E_A , the electrons follow the flux adiabatically. To leading order in $\hbar\omega/E_A$, Eq. (1) reduces to a single-particle problem for the dimensionless flux $X = a + a^\dagger$,

$$H_{\text{BO}} = \frac{\hbar\omega}{4} (P^2 + X^2) + E(\varphi_{\text{cl}} + gX), \quad [X, P] = 2i. \quad (3)$$

Equation (3) is the Hamiltonian of an rf-SQUID [3] whose inductive energy is supplied by the mode. Writing it in the Cooper-pair phase $\theta = 2gX$ gives $4E_C n^2 + \frac{1}{2}E_L \theta^2 + E(\theta/2)$ with $E_L = \hbar\omega/8g^2$ and $E_C = \hbar\omega g^2$. A static field ($\langle a \rangle \neq 0$), i.e. a photon condensate carrying a persistent current $I = -(2e/\hbar)\partial_\varphi E$, appears classically when the orbital response of the ring at the symmetric point is paramagnetic and exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2 |E''(\varphi_0)|}{\hbar\omega} > 1, \quad E''(\varphi_0) < 0. \quad (4)$$

For a pure π -junction, $\beta = 8g^2 E_J / \hbar\omega = 2\pi L I_c / \Phi_0$, the familiar screening parameter of the loop [4, 5]. Equation (4) is the single-loop form of the photon-condensation criterion of Andolina *et al.* [6]: the no-go theorem for a uniform cavity field [7, 8] does not apply, because the mode couples to the flux (curl of $\mathbf{A}$), and the instability requires a *paramagnetic* orbital response, which here comes from the π shift. Above threshold, the Landau free energy $\mathcal{F}(X) = \hbar\omega X^2/4 + E(gX)$ has two degenerate minima $\pm X^*$ carrying opposite currents $\pm I^*$.

Two remarks fix the physical content of Eq. (4). First, for a loop with its own geometric inductance L_r and

mutual inductance $M \leq \sqrt{L_r L_c}$ to the resonator, minimizing the static energy over the resonator flux gives $\Phi_r^2/2L_r$: the static threshold is set by the total loop inductance, and Eq. (1) is the $k = M/\sqrt{L_r L_c} \rightarrow 1$ (galvanic) limit. Second, zero-point fluctuations of the mode do not drive the transition; they oppose it. At finite $E_J/\hbar\omega$ the ground state of Eq. (3) is the symmetric superposition of the two current-carrying states, split from its partner by a tunnelling gap $\delta \sim \hbar\omega e^{-S}$ with an instanton action $S \propto (E_J/\hbar\omega)s(\beta)$. A sharp quantum phase transition with mean-field exponents is recovered in the scaling limit $E_J/\hbar\omega \rightarrow \infty$ at fixed β , in which $E_C/E_J = \beta(\hbar\omega)^2/(8E_J^2) \rightarrow 0$. This is the analogue of the classical-oscillator limit of the quantum Rabi model [9].

Methods. We solve Eq. (1) without approximation by DMRG (TeNPy [10]) on the full electron–photon Hilbert space. The ring is stored in folded order next to a bosonic site (Fock cutoff N_{ph}). Fermion parity and S^z are conserved, and the electron–photon vertex is a three-site operator $D c_{1\sigma}^\dagger c_{0\sigma}$. Benchmarks against exact BdG energies of the non-interacting ring at $g = 0$ agree to 10^{-14} at full bond dimension ($\chi = 1024$). In production ($\chi = 256\text{--}320$, $N_{\text{ph}} = 30\text{--}50$, discarded Fock weight $< 10^{-7}$) the ground-state energy error is $\lesssim 3 \times 10^{-6}t$. For a 4-site ring with a photon, DMRG reproduces exact diagonalization and the identity $\langle I \rangle = \hbar\omega \langle X \rangle / 2g$ to 10^{-7} . Excited states are obtained by DMRG orthogonal to the ground state. We compare with (i) the self-consistent electron×photon product state (mean field, MF), and (ii) the adiabatic Hamiltonian (3), with $E(\varphi)$ from BdG or from DMRG at fixed φ , diagonalized exactly.

Spontaneous persistent current in the microscopic model. Figure 2 contains the central result. Panel (a) follows the ground state of Eq. (1) at $U = 2t$ as the coupling g is increased. For $g < g_c \simeq 0.12$ the mode stays close to its vacuum ($\langle a^\dagger a \rangle \lesssim 1$) and the current is the linear response to the infinitesimal seed flux, $|\langle I \rangle| \lesssim 10^{-4}et/\hbar$. At $g \simeq g_c$ the photon number jumps to $\langle a^\dagger a \rangle \simeq 8$ and a persistent current $|\langle I \rangle| \simeq 1.4 \times 10^{-2}et/\hbar \simeq 0.93 I_c$ appears. Reducing the seed by two orders of magnitude leaves the current above threshold unchanged, while the current below threshold scales to zero with the seed: the current is spontaneous. Full DMRG of the electron–photon problem (symbols) and the exact treatment of the photon in the adiabatic Hamiltonian (3) with the DMRG $E(\varphi)$ (lines) agree within a few percent. Beyond the maximum the current decreases as $I \simeq \hbar\omega\varphi^*/2g^2$, because the condensate flux saturates at $\Phi_0/2$ while the effective inductance $\propto g^2/\hbar\omega$ grows.

Figure 2(b) shows the current in the (U, g) plane at fixed $\hbar\omega = 7.6 \times 10^{-4}t$. Its boundary has two parts. The lower edge, $U_c \simeq 0.9t$, is the $0\text{--}\pi$ transition of the junction itself. For $U < U_c$ the ground state is a singlet with $E_2 < 0$, the orbital response is diamagnetic, and no current appears at any g . The left edge, $g_c(U)$, follows the classical criterion $\beta(U, g) = 1$ and moves to stronger

coupling as $E_2(U)$ decreases. Panel (c) is the same map from the mean-field theory of Ref. [1]: Hartree–Fock BdG electrons times a photon product state, with the same seed. For $U \lesssim 5t$, i.e. $E_J/\hbar\omega \gtrsim 3$, mean field reproduces U_c and $g_c(U)$ to within the grid resolution ($\delta g = 0.01$). It underestimates the maximal current by 10–20% (the product state renormalizes the junction bond by $|\langle D \rangle| < 1$). For $U \gtrsim 6t$, where E_2 drops to $E_J \lesssim 3\hbar\omega$, mean field predicts an ordered phase (dashed line in (b)) that is absent in the exact solution. Quantum fluctuations of the flux restore the symmetry there, and the ground state remains the cat state described below. In short, the product state is reliable only when $E_J \gg \hbar\omega$ and $g \ll 1$. Both conditions are violated by the parameters of the original notebook, see Fig. 3: at $\hbar\omega = 0.02$ the exact ground state carries no current at any g , whereas at $\hbar\omega = 0.002$ the same microscopic model orders at $g_c \simeq 0.21$, while mean field predicts $g_c \simeq 0.30$.

Breakdown of the product-state description for $E_J \simeq \hbar\omega$. We first take the parameters of a Zeeman-polarized dot embedded in a short BCS ring ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$). The self-consistent product state predicts a current-carrying phase for $g \gtrsim 0.75$, with $\langle a^\dagger a \rangle \simeq 1\text{--}1.4$ and $|\langle I \rangle| \approx 0.02$ [Fig. 5]. DMRG shows that this phase is an artefact. The exact ground-state energy decreases monotonically with g and lies below the MF energy by up to $1.8\hbar\omega$ (at $g = 1.2$). The gap to the first excited state stays a sizeable fraction of $\hbar\omega$ ($E_1 - E_0 = 0.33\hbar\omega$ at $g = 1.2$), and the photon number saturates at $\langle a^\dagger a \rangle \simeq 0.36$. The response to the seed flux $\varphi_{\text{cl}} = 10^{-3}$ is linear: $\langle I \rangle \approx 7 \times 10^{-5}$, more than two orders of magnitude below the MF value. Here and below we evaluate the current through the exact relation $\langle I \rangle = \hbar\omega \langle X \rangle / 2g$, which follows from $\langle [H, P] \rangle = 0$ and states that the persistent current is the static displacement of the mode. DMRG and the adiabatic theory (3) agree to within $0.06\hbar\omega$ in energy and 1.3% in $E_1 - E_0$. The residual difference is the non-adiabatic (diagonal Born–Oppenheimer) correction, of relative order $\hbar\omega/E_A \approx 0.1$.

Two effects make the product state fail. (i) It is not gauge invariant [11, 12]: with the full Peierls phase on one bond, it renormalizes that bond by $|\langle D \rangle| < 1$ and so loses phase-independent kinetic energy, which the exact state recovers by following the flux adiabatically. (ii) Here $E_J \simeq \hbar\omega$. The Landau free energy does develop a double well for $g > 0.58$, but its depth is below the zero-point energy, and the flux distribution $P(X)$ stays essentially unimodal. A further subtlety concerns ring size: since $L \lesssim \xi$, $E(\varphi)$ of this ring is dominated by the single-electron harmonic, $E_1 = 2.1 \times 10^{-2} \gg E_2 = 2.0 \times 10^{-3}$ [Fig. 4(a)]. The instability would then be one of the normal h/e persistent current rather than of the Josephson junction. To isolate Josephson physics we use rings with $L \gg \xi$ below ($t = 1$, $\Delta = 1$, $L = 14$, giving $|E_1/E_2| \lesssim 10^{-2}$).

Half-flux-biased junction. For an ordinary weak link ($t_J = 0.5$, $E_2 = -0.056t$, Andreev gap $0.77t$) biased by $\Phi_0/2$, Fig. 6 shows the full DMRG solution at $E_J/\hbar\omega = 10$ as a function of β . The current rises steeply at $\beta \simeq 1.1$ and saturates at $|\langle I \rangle| \simeq 0.97I_c$ for $\beta \approx 1.7$. It then decreases as the flux approaches $\Phi_0/2$, where the junction current vanishes. DMRG and the adiabatic theory agree to $0.003\hbar\omega$ in energy, to $\lesssim 5\%$ in current and to 2% in $E_1 - E_0$ at all β . The product state is $0.2\text{--}0.4\hbar\omega$ too high in energy and overestimates the current threefold near threshold ($0.71I_c$ versus $0.21I_c$ at $\beta = 1.13$). It becomes accurate only deep in the ordered phase, where the photon is a nearly coherent state. The flux distribution extracted from the photon reduced density matrix evolves from a Gaussian centred at zero to a single displaced peak at $\pm\Phi^*$. The sign is fixed by the residual odd harmonic $E_1 = -5.4 \times 10^{-4}$, which tilts the double well by $2|E_1|\sin\varphi^* \approx 10^{-3}t \approx 0.2\hbar\omega$. This tilt sets the splitting $E_1 - E_0$ for $\beta > 1.2$ in DMRG and BO alike. Without it, $\delta/\hbar\omega$ would drop below 10^{-6} at $\beta = 2$. The external-bias route thus provides a current-carrying state whose direction is *selected* rather than spontaneously chosen. Its residual symmetry breaking is $\propto e^{-L/\xi}$ for a homogeneous ring, but in practice it is limited by the flux offset $\delta\Phi$.

Interaction-driven π junction without external flux. We now replace the weak link by an Anderson impurity at particle-hole symmetry ($\epsilon_d = -U/2$, $t_J = 0.5$) and set $\varphi_{cl} = 0$. DMRG in the even and odd fermion-parity sectors [Fig. 4(b,c)] gives the $0\text{--}\pi$ transition known for magnetic quantum dots [13–19]. For $U < U_c \simeq 0.9t$ the singlet (even) state is the ground state at $\varphi = 0$ and the junction is a 0 -junction. For $U > U_c$ the Kramers doublet (odd sector) is the ground state at all phases, and $E(\varphi)$ has its *minimum* at the Cooper-pair phase π . At $U = 2t$ we find $E_2 = +7.39 \times 10^{-3}t$, with $|E_1|, |E_4| < 10^{-4}t$: a pure π junction, indistinguishable after normalization from the half-flux-biased weak link [Fig. 4(a)]. The π coupling decreases monotonically with U ($E_2 \propto U^{-1.3}$ for $4t \leq U \leq 10t$), as expected for cotunnelling through the singly occupied level.

Coupling this junction to the mode [Fig. 7] gives the transition without any static flux. Here H_{ring} is real and $\langle I \rangle = 0$ in every eigenstate, so the order is diagnosed by the two lowest states. The splitting $\delta = E_1 - E_0$ collapses and the flux operator projected onto the quasi-degenerate doublet acquires the eigenvalues $\pm\Phi^*$. At $E_J/\hbar\omega = 10$, DMRG and the adiabatic theory give the same Φ^* to within 3% for all β [e.g. $\Phi^* = 0.57\Phi_0/2$ at $\beta = 2$]. They also agree on $\delta/\hbar\omega = 0.71$ and 0.28 at $\beta = 0.5$ and 1 . Deeper in the ordered phase, δ drops below the DMRG resolution for excited-state energies ($\sim 3 \times 10^{-5}t \approx 0.04\hbar\omega$, estimated by comparing $\chi = 256$ and 400 at $\beta = 1.5$). The adiabatic theory, validated above, then gives $\delta/\hbar\omega = 7 \times 10^{-4}$, 2×10^{-6} and 2×10^{-8} at $\beta = 1.5$, 2 and 3 . In the photon-number

basis the DMRG ground state is a cat state: $P(\Phi)$ is bimodal with peaks at $\pm\Phi^*$, whose separation tracks the classical minimum. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [Fig. 7(b)]. The residual δ is the macroscopic tunnelling of a persistent current. Because it is protected by time-reversal symmetry, it is the only energy scale separating the two current directions. Each current state is additionally Kramers degenerate in the spin of the dot. The low-energy manifold is therefore a fourfold multiplet $\{\sigma = \uparrow, \downarrow\} \otimes \{\odot, \ominus\}$, split only by δ .

Experimental scales. The two control parameters are $\beta = 2\pi LI_c/\Phi_0$ and $E_J/\hbar\omega$, related by $\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$, with $E_J/h \simeq 0.5\text{ GHz} \times I_c[\text{nA}]$. Three classes of π elements span the phase diagram [Fig. 1(b)]: (i) SFS junctions ($I_c \sim 1\text{--}100\mu\text{A}$) reach $\beta > 1$ with geometric inductances of a few to a few hundred pH and $E_J/\hbar\omega \sim 10^2\text{--}10^4$. This is the classical limit, where the half-flux state is fully polarized, consistent with the spontaneous currents observed in SFS and d -wave π loops [5, 20, 21]. (ii) Quantum-dot π junctions in the doublet regime ($I_c^\pi \sim 0.1\text{--}10\text{ nA}$) require $L \sim 0.03\text{--}3\mu\text{H}$, i.e. superinductances (Josephson-junction arrays, granular Al) as in fluxonium circuits [22, 23]; this is the π -qubit regime of Refs. [24–26]. These give $E_J/\hbar\omega \sim 1$: the ground state is the cat state $(|\odot\rangle + |\ominus\rangle)/\sqrt{2}$ with $\delta \sim 0.1\hbar\omega$, and the transition appears as a dip in the $|0\rangle \rightarrow |1\rangle$ frequency and a jump in the transition matrix element $\langle 0|\Phi|1\rangle$, accessible in two-tone spectroscopy. (iii) Multichannel or gate-tunable π junctions with $I_c \sim 50\text{--}200\text{ nA}$ in a high-kinetic-inductance loop ($L \sim 2\text{--}10\text{ nH}$, $\omega/2\pi \sim 2\text{--}5\text{ GHz}$) give $\beta \sim 1\text{--}3$ and $E_J/\hbar\omega \sim 10\text{--}30$. This is the regime of Figs. 6 and 7: a well-defined current-carrying state whose residual tunnelling $\delta/\hbar\omega \sim 10^{-4}\text{--}10^{-8}$ remains a measurable macroscopic-quantum-tunnelling signature. In the externally biased realization, the half-flux bias must be stable to $|\delta\Phi| \ll \delta/2I^*$, i.e. $\lesssim 10^{-6}\Phi_0$ for the parameters above. The interaction-driven realization has no such requirement.

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- [1] Mean-field notebook `Optimised_SC_ring_with_JJ_cavity_mean_field` (this work, earlier stage).
 - [2] D. Loss and P. Goldbart, Phys. Rev. B **43**, 13762 (1991).
 - [3] J. R. Friedman, V. Patel, W. Chen, S. K. Tolpygo, and J. E. Lukens, Nature **406**, 43 (2000).
 - [4] L. N. Bulaevskii, V. V. Kuzii, and A. A. Sobyanin, JETP Lett. **25**, 290 (1977).
 - [5] C. C. Tsuei and J. R. Kirtley, Rev. Mod. Phys. **72**, 969 (2000).
 - [6] G. M. Andolina, F. M. D. Pellegrino, V. Giovannetti, A. H. MacDonald, and M. Polini, Phys. Rev. B **102**, 125137 (2020).
 - [7] G. M. Andolina, F. M. D. Pellegrino, V. Giovannetti, A. H. MacDonald, and M. Polini, Phys. Rev. B **100**, 121109(R) (2019).

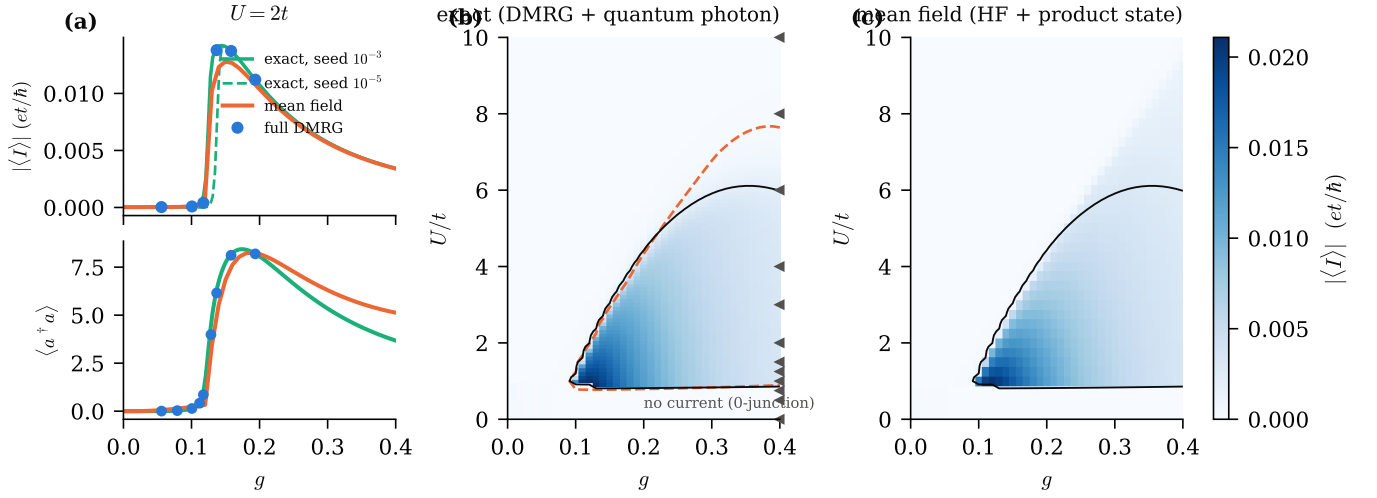


FIG. 2. Spontaneous persistent current in the microscopic model Eq. (1) ($L = 14$, $t = 1$, $\Delta = t$, $t_J = 0.5t$, $\epsilon_d = -U/2$, $\hbar\omega = 7.6 \times 10^{-4}t$, no external flux). (a) Current and photon number versus g at $U = 2t$: full DMRG (circles), exact photon in the adiabatic Hamiltonian with DMRG $E(\varphi)$ (green; solid and dashed: seeds $\varphi_{cl} = 10^{-3}$ and 10^{-5}), and mean field (orange). (b) Exact current in the (U, g) plane (seed 10^{-3}); black line: $|\langle I \rangle| = 2 \times 10^{-3} et/\hbar$; orange dashed: the same contour from mean field; arrows: U values computed by DMRG. (c) Mean field (Hartree-Fock BdG $\times$ photon product state), same color scale.

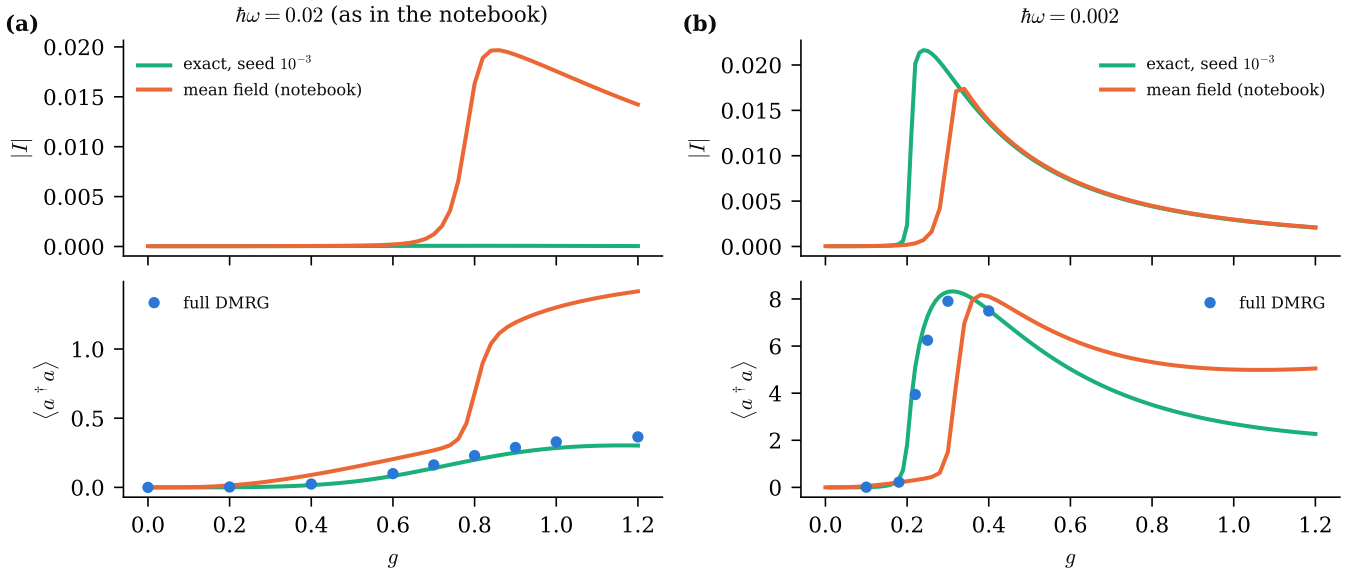


FIG. 3. Model of the original notebook (Zeeman-polarized dot, $L = 11$, $t = 2$, $\Delta = 0.6$): current (top) and photon number (bottom) versus g for (a) $\hbar\omega = 0.02$ and (b) $\hbar\omega = 0.002$. Green: exact (adiabatic, seed 10^{-3}); orange: mean field; circles: full DMRG.

- [8] P. Nataf, T. Champel, G. Blatter, and D. M. Basko, Phys. Rev. Lett. **123**, 207402 (2019).
- [9] M.-J. Hwang, R. Puebla, and M. B. Plenio, Phys. Rev. Lett. **115**, 180404 (2015).
- [10] J. Hauschild and F. Pollmann, SciPost Phys. Lect. Notes, **5** (2018).
- [11] J. Li, D. Golez, G. Mazza, A. J. Millis, A. Georges, and M. Eckstein, Phys. Rev. B **101**, 205140 (2020).
- [12] O. Dmytruk and M. Schiró, Phys. Rev. B **103**, 075131 (2021).
- [13] L. I. Glazman and K. A. Matveev, JETP Lett. **49**, 659 (1989).
- [14] B. I. Spivak and S. A. Kivelson, Phys. Rev. B **43**, 3740 (1991).
- [15] E. Vecino, A. Martín-Rodero, and A. L. Yeyati, Phys. Rev. B **68**, 035105 (2003).
- [16] C. Karrasch, A. Oguri, and V. Meden, Phys. Rev. B **77**, 024517 (2008).
- [17] J. A. van Dam, Y. V. Nazarov, E. P. A. M. Bakkers, S. D. Franceschi, and L. P. Kouwenhoven, Nature **442**, 667 (2006).

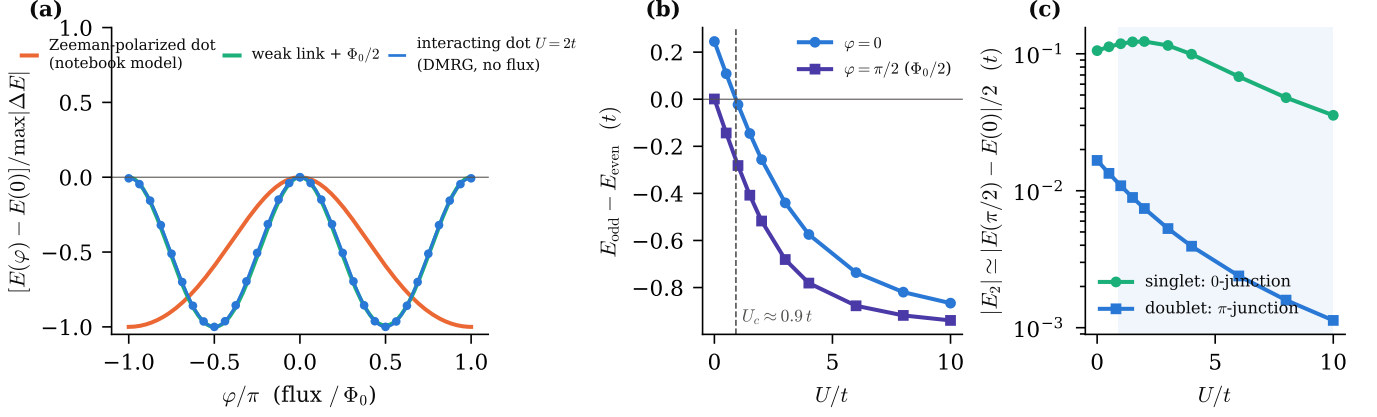


FIG. 4. (a) Normalized ground-state energy of the loop versus the electron phase φ (flux in units of $h/2e$) for three junctions: Zeeman-polarized dot in a short ring (dominated by the single-electron harmonic $\cos \varphi$), an ordinary weak link biased by $\Phi_0/2$, and an interacting dot at $U = 2t$ without any bias (DMRG, symbols; line: cosine fit). (b) Parity competition $E_{\text{odd}} - E_{\text{even}}$ of the interacting-dot ring at $\varphi = 0$ and $\varphi = \pi/2$ (DMRG, $L = 14$, $\Delta = t$, $t_J = 0.5t$). (c) Josephson energy $|E_2|$ in the singlet (0) and doublet (π) sectors. Shading: doublet ground state.

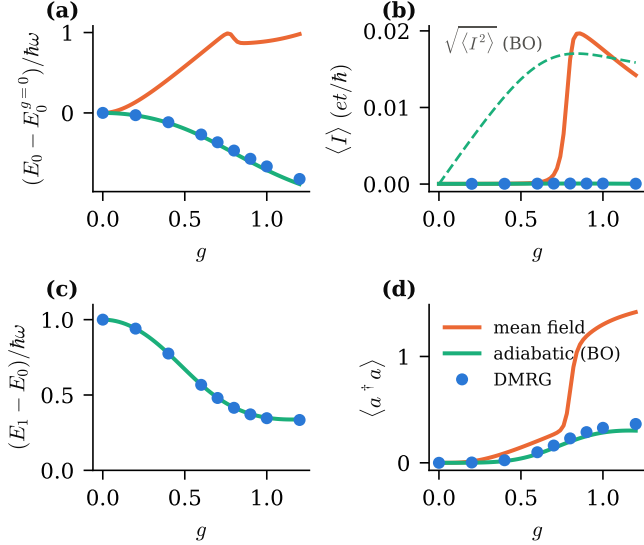


FIG. 5. Short ring with a Zeeman-polarized dot ($\hbar\omega = 0.02$, $\varphi_{\text{cl}} = 10^{-3}$): mean field (MF, orange), adiabatic theory (3) (green), and DMRG (symbols). (a) Ground-state energy, (b) current (dashed: rms current of the BO ground state), (c) gap to the first excited state, (d) photon number.

- [18] J.-P. Cleuziou *et al.*, Nat. Nanotechnol. **1**, 53 (2006).
- [19] R. Delagr ange *et al.*, Phys. Rev. B **91**, 241401(R) (2015).
- [20] V. V. Ryazanov *et al.*, Phys. Rev. Lett. **86**, 2427 (2001).
- [21] A. Bauer *et al.*, Phys. Rev. Lett. **92**, 217001 (2004).
- [22] V. E. Manucharyan, J. Koch, L. I. Glazman, and M. H. Devoret, Science **326**, 113 (2009).
- [23] L. Gr nhaupt *et al.*, Nat. Mater. **18**, 816 (2019).
- [24] L. B. Ioffe, V. B. Geshkenbein, M. V. Feigel'man, A. L. Fauch re, and G. Blatter, Nature **398**, 679 (1999).
- [25] T. Yamashita, K. Tanikawa, S. Takahashi, and S. Maekawa, Phys. Rev. Lett. **95**, 097001 (2005).
- [26] A. K. Feofanov *et al.*, Nat. Phys. **6**, 593 (2010).

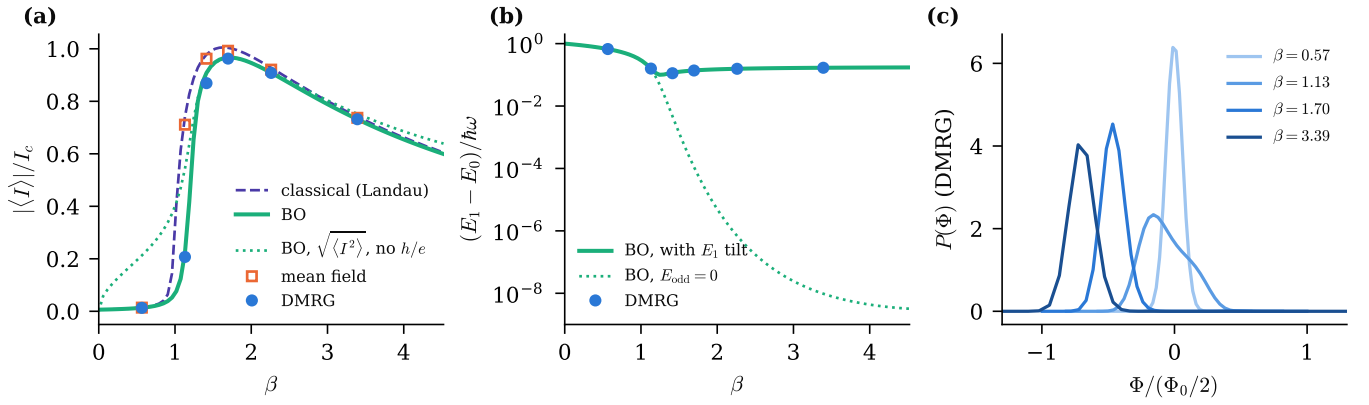


FIG. 6. Weak-link loop biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current versus β : DMRG (circles), MF (squares), adiabatic theory with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state, obtained from the photon reduced density matrix.

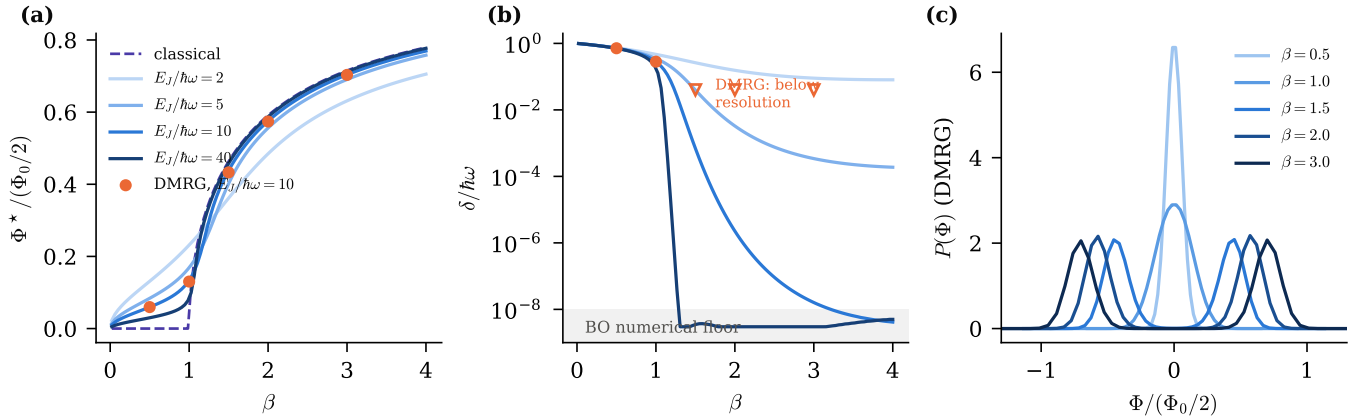


FIG. 7. Interaction-driven π loop ($U = 2t$, no external flux). (a) Flux of the condensate Φ^* (eigenvalue of the flux projected on the two lowest states) versus β for increasing $E_J/\hbar\omega$ (lines: adiabatic theory with the DMRG $E(\varphi)$; circles: full DMRG at $E_J/\hbar\omega = 10$; dashed: classical). (b) Tunnel splitting δ between the two current-carrying states. (c) Flux distribution of the DMRG ground state (a cat state for $\beta > 1$).